BYOUNGUK MIN

Department of Computer Science, Purdue University
Indiana, United States 47907
+01-7654097422 ⋄ min87@purdue.edu

EDUCATION

Purdue University *Aug. 2021–Present*
PhD student in Computer Science
Advisor: Professor Changhee Jung

Yonsei University *Mar. 2019–Feb. 2021*
Master student in Computer Science
Advisor: Professor Bernd Burgstaller

Sungkyunkwan University *Mar. 2013–Feb. 2019*
BS in Computer Science

PUBLICATION

- "iGPU: Enabling GPUs for Intermittent AI on Batteryless Systems" by **Byounguk Min**, Yilun Wu, Y. Zhou, Kasım Sinan Yıldırım, Khakim Akhunov, Jianping Zeng, Jongouk Choi, Wenjie Xiong, Dongyoon Lee, and Changhee Jung. In Proceedings of the 59th IEEE/ACM International Symposium on Microarchitecture, MICRO'26, Athens, Greece, 2026.
- "Achieving Speculative Intermittent Computation on Commodity Energy Harvesting Hardware" by Yida Zhang, Shao-Yu Huang, **Byounguk Min**, Andrew Inho Park, Gan Fang, Jongouk Choi, and Changhee Jung.
- "WarmCache: Exploiting STT-RAM Cache for Low-Power Intermittent Systems" by Noureldin Hassan, **Byounguk Min**, Changhee Jung, Yan Solihin, and Jongouk Choi. In Proceedings of the 52nd ACM/IEEE International Symposium on Computer Architecture, ISCA'25, Tokyo, Japan, 2025.
- "IntOS: Persistent Embedded Operating System and Language Support for Multi-threaded Intermittent Computing" by Yilun Wu, **Byounguk Min**, Mohannad Ismail, Wenjie Xiong, Changhee Jung, and Dongyoon Lee. In Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation, OSDI'24, Santa Clara, CA, USA, 2024.

RESEARCH PROJECTS

Enabling Heterogeneous Computing System Purdue University
Supervised by Professor Changhee Jung *Dec. 2025–Present*

· Developing a power-aware work-partitioning scheme that splits each inference task across engines according to the available energy budget, with idempotent recovery when work is reassigned after a power failure.

LEAF: Equipping Nonvolatile Processors with Volatile Caches on the Cheap Purdue University
Supervised by Professor Changhee Jung *Dec. 2025–Present*

· Designed LEAF, a hardware-only store-logging and replay mechanism that enables volatile L1 data caches in nonvolatile processors without cacheline checkpointing.
· Implemented asynchronous persistence with ring-buffer log management to reduce stalls, backup energy, and area overhead.

IC-SIM: Intermittent Computing Simulator for Energy Harvesting System Purdue University
Supervised by Professor Changhee Jung *Nov. 2025–Present*

· This project proposes/builds IC-SIM, an intermittent computing simulator for energy-harvesting systems that models power failures, capacitor/battery dynamics, and checkpoint/restart behavior to study end-to-end progress under intermittent power.

Enabling non-volatile GPU for Energy Harvesting System Purdue University
Supervised by Professor Changhee Jung *Nov. 2022–Aug. 2026*

· Although numerous studies have investigated CPU design exploration for energy harvesting systems; however, research on GPU architectures in this domain remains limited. This project aims to design a GPU architecture optimized for energy harvesting systems.

Exploiting STT-RAM Cache for intermittent system Purdue University
Supervised by Professor Changhee Jung *Nov. 2024–June. 2025*

· This project is to develop WarmCache, optimized STT-RAM cache design with relaxed non-volatility, for power-hungry energy harvesting system.

Embedded operating system for energy-harvesting platform Purdue University
Supervised by Professor Changhee Jung *Oct. 2023–Mar. 2024*

· This project is to develop IntOS, an embedded operating system and language support for multi-threaded intermittent computing on a battery-less energy-harvesting platform
· IntOS simplifies programming with a traditional "thread" and a "transaction" with automatic undo-logging of persistent objects in non-volatile memory.

Persistent Lock-Free Queue with Batching for Non-volatile Memory Yonsei University
Supervised by Professor Bernd Burgstaller. *Jan. 2020–June. 2020*

· Presented PBQ, a novel persistent lock-free queue using bathch mechanism to provide lock-free queue with high performance and durability under non-volatile memory.
· Experimented with DCPM server and manycore server (120 cores).

Evaluation of relaxed memory consistency model under RISC-V Yonsei University
Supervised by Professor Bernd Burgstaller. *Jan. 2020–Apr. 2020*

· Extending the previous work, evaluated non-blocking concurrent object with atomic read-modify-write operations supported by RISC-V architecture.
· Experimented under Hifive Unleashed which has 4 physical cores and test the scalability of the non-blocking algorithms.

Design of non-blocking synchronization in Ada programming language Yonsei University
Supervised by Professor Bernd Burgstaller *Apr. 2019–Nov. 2019*

· Proposed concurrent objects as novel programming primitive to encapsulate the complexity of non-blocking synchronization and exploit atomic read-modify-write operations under relaxed memory consistency model.
· Experimented under x86 and ARMv8 architecture to monitor how non-blocking concurrent objects are affected by the different memory consistency models.
· Published in Journal of Systems Architecture.

TEACHING ASSISTANCE

Compiling And Programming Systems(CS50200) Purdue University
Led by Professor Changhee Jung Jan. 2024– May. 2024, Jan. 2025–May. 2025, Jan. 2026–May. 2026

· Covered advanced topics of compiler design for graduated students.

Compilers: Principles and Practice(CS35200) Purdue University
Led by Professor Changhee Jung Aug. 2023–Dec. 2023, Aug. 2024–Dec. 2024, Aug. 2025–Dec. 2025, Aug. 2026–Present

· Covered fundamental topics of compiler design for undergraduate students.

Computer Programming(CSI2100-01) Yonsei University
Led by Professor Bernd Burgstaller. Mar. 2019–July 2019, Mar. 2020–July 2020

· Covered fundamentals of Python programming language and programming techniques.

Compiler Design(CSI4104-01) Yonsei University
Led by Professor Bernd Burgstaller Sept. 2019–Dec. 2019

· Covered programming language design patterns and compiler techniques.

TECHNICAL STRENGTHS

- Proficient in C/C++, Python programming language.
- Proficient in gem5 simulator, which is open-source system-level and processor simulator.
- Proficient in GPGPU-Sim (AccelSim), an open-source system-level and processor simulator for GPUs.
- Proficient in LLVM compiler infrastructure.
- Proficient in parallel and concurrent programming.
- Experienced in programming on non-volatile memory and PMDK library.

EXTRACURRICULAR ACTIVITIES

- Serving in the USFK(United States Forces Korea) as a KATUSA(Korea Augmentation To the United States Army) *June 2015–Mar. 2017*